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Department Information Technology
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Idea Lab Presentation on

" Smart and Sustainable School Transportation System for Mira Bhayandar. "

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Contents :-

- Problem statement
- Abstract
- Motivation
- Objectives
- Introduction
- Literature Survey
- Features of Proposed System
- Hardware and Software Requirements
- Proposed Architecture
- Expected Results and Discussion
- Conclusion
- References

Problem statement

Many school transportation systems in Mira Bhayandar operate without real-time vehicle tracking or route optimization. Parents have no live updates on bus locations, and inefficient bus routing increases fuel consumption, commute time, and traffic congestion near schools. There is a need to improve safety, reduce carbon footprint, and streamline school logistics

Abstract

The proposed Smart School Transportation & Tracking Platform is a ***software-only, AI-driven system*** designed to provide real-time bus visibility and route optimization using ***driver smartphones instead of GPS hardware***. The platform integrates AI scheduling, WhatsApp-based tracking, and safety compliance monitoring to reduce congestion, fuel consumption, and operational costs while improving student safety and urban mobility in Mira–Bhayandar.

Motivation

Current school transportation in Mira Bhayandar lacks real-time tracking and efficient route planning, causing safety concerns, delays, fuel wastage, and traffic congestion near schools. Parents often remain uncertain about bus locations, while schools struggle with coordination and rising operational costs. These challenges highlight the need for a smarter and more environmentally responsible solution. The motivation behind this project is to enhance student safety, provide transparency to parents, reduce carbon footprint, and improve overall transport efficiency through ***sustainable innovative development*** and technology-driven logistics.

Objectives

- To design a real-time school bus tracking system that allows parents and schools to monitor vehicle location instantly.
- To optimize bus routes in order to reduce travel time, fuel consumption, and traffic congestion near school areas.
- To improve student safety through timely alerts, clear communication, and better transport coordination.
- To lower environmental impact by promoting fuel-efficient and eco-friendly transportation practices.
- To support ***sustainable innovative development*** by integrating practical technology solutions into everyday school logistics.

Introduction

School transportation in Mira Bhayandar has become increasingly challenging due to growing traffic, lack of real-time vehicle tracking, and inefficient route planning. These issues lead to delays, safety concerns, poor communication with parents, and higher fuel consumption, which also contributes to congestion near school areas. There is a clear need for a smarter and more responsible system that improves transparency, efficiency, and student safety. By applying technology-based monitoring and route optimization with a focus on ***sustainable innovative development***, school logistics can become more reliable, eco-friendly, and better suited for future urban growth.

Literature survey

Sr. No.	Name of Document	Author/ Organization	Abstract
1	Logistics in Lean Systems: Transport Impact	V. Kujanpää (Aalto University, 2024)	Case studies prove JIT scheduling reduces wait times by 40% and improves transit performance by 35%. Validates the use of WhatsApp-based scheduling to achieve a 75% on-time target.
2	Urban Transport Governing Guidelines	M. Janné et al. (2019)	Analysis shows coordinated logistics reduce small vehicle trips 70% and improve traffic flow 45%. Proves that municipal coordination (MBMC model) is critical for a 35% reduction in road blockages.
3	Smart Mobility for Urban Sustainability	P. Haag et al. (Frontiers, 2023)	Urban transit faces 2.5x more logistics challenges, but Smartphone GPS solves 65% of these constraints. A software-only approach is perfect for Tier-2 city pilots like Mira- Bhayandar.
4	Digital Supply Chain Optimization - Indian Cities	Deloitte Global (2025)	Study of 15 cities shows AI scheduling delivers 32% efficiency within 6 months and a 4.2-month ROI. Confirms that WhatsApp reaches 95% of drivers, making it an exact match for a practical MVP.
5	Mira-Bhayandar Urban Mobility Baseline	MBMC & IIT Bombay (2025)	Confirms that uncoordinated transport contributes to 27% of peak congestion. Estimates a ₹15–20 Cr annual economic loss, with a potential ₹5 Cr value created from a small ₹8 L investment.

Features of Proposed System

- ***AI Route Optimization Engine***
- ***Smartphone GPS / WhatsApp Tracking***
- Parent ETA Notifications
- Safety & Driver Compliance Monitoring
- School Management Dashboard
- ***Zero Hardware Dependency***

Hardware and Software Requirements

1. Hardware

- Driver Smartphones
- School Admin Computer/Laptop
- Internet Connectivity

2. Software

- React Web Dashboard
- Node.js Backend
- PostgreSQL Database
- Google Maps API
- WhatsApp Business API
- AWS Cloud Hosting

Proposed Architecture

A ***cloud-centric architecture*** where driver smartphones transmit GPS data to a central server. AI algorithms process routing and scheduling, while schools monitor fleets via a dashboard. Parents receive live ETAs through messaging services. The system is designed for ***real-time concurrency, scalability, and low infrastructure cost***.

Expected Results and Discussion

- ***75% improvement*** in punctuality
- ***35% reduction*** in congestion near schools
- ***100% live visibility*** for parents
- Noticeable drop in fuel consumption and emissions
- ***85% adoption rate*** among drivers and parents
- Potential ***₹5 crore economic value creation annually***

Conclusion

The platform introduces a ***practical, scalable, and financially sustainable*** approach to school transportation management. By eliminating hardware dependency and leveraging AI with smartphone technology, the system enhances safety, reduces urban congestion, and promotes environmental responsibility while remaining affordable for schools in Tier-2 cities.

References

- Urban Mobility and Logistics Research Papers
- AI Transportation Optimization Studies
- Smartphone GPS Adoption Reports
- Mira–Bhayandar Municipal Traffic Baseline Studies
- Sustainable Transport Technology Journals